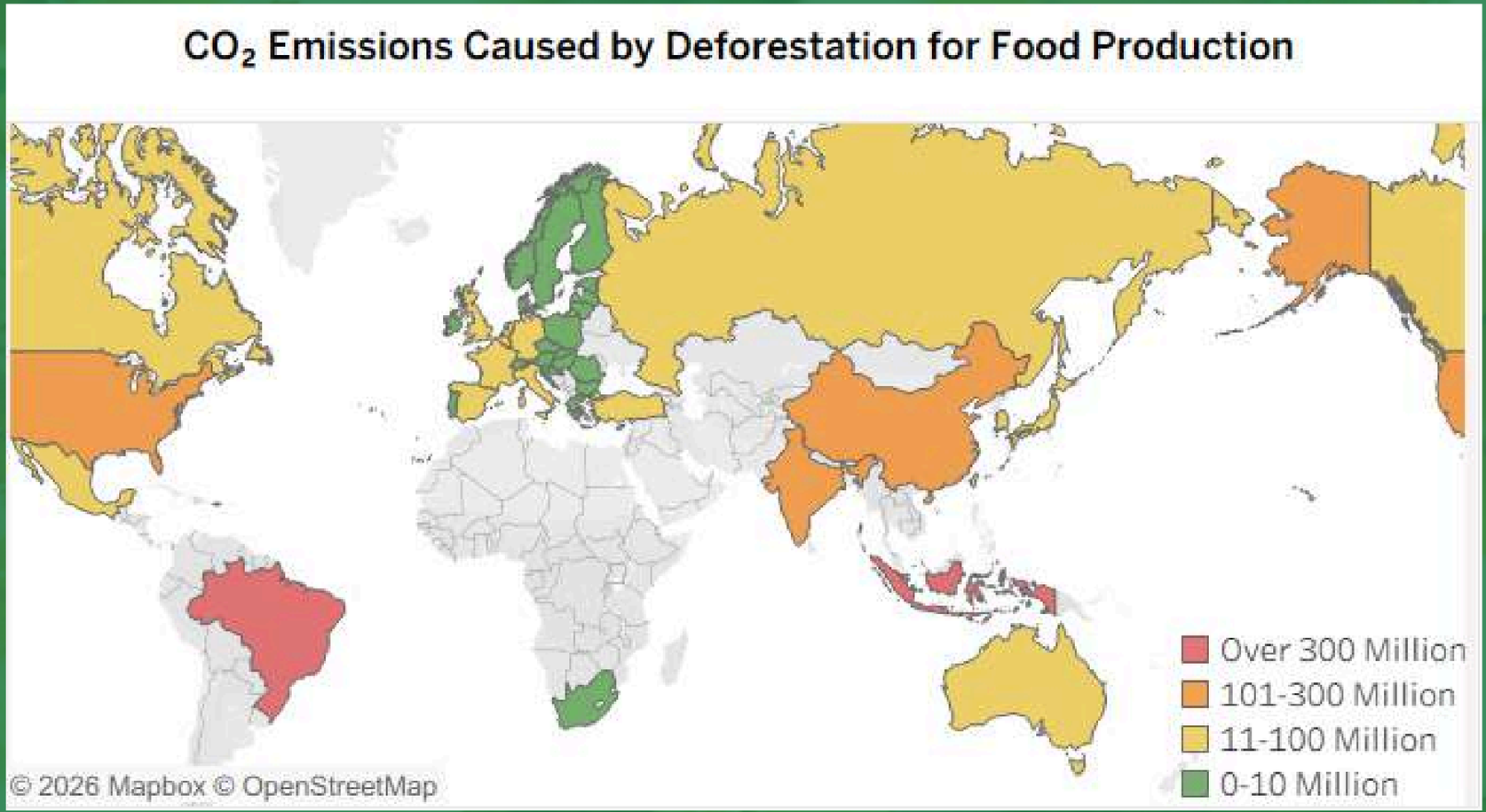
DESIGNING A LOW-CARBON MENU: DATA-DRIVEN INSIGHTS FOR A SUSTAINABLE RESTAURANT

Why what we eat matters more than where food comes from

- Data: GHG Emissions by Lifecycle Stage (Poore & Nemecek, Science 2018)
- https://ourworldindata.org

FOOD AND CLIMATE CHANGE



Food production is responsible for around 25% of global greenhouse gas emissions.
Globally, 8 out of 10 people consider climate change a major threat to their country.

THE “EAT LOCAL” MYTH

A common recommendation is to “eat local” to reduce emissions.

However, data shows that transport represents only a small share of food-related emissions.

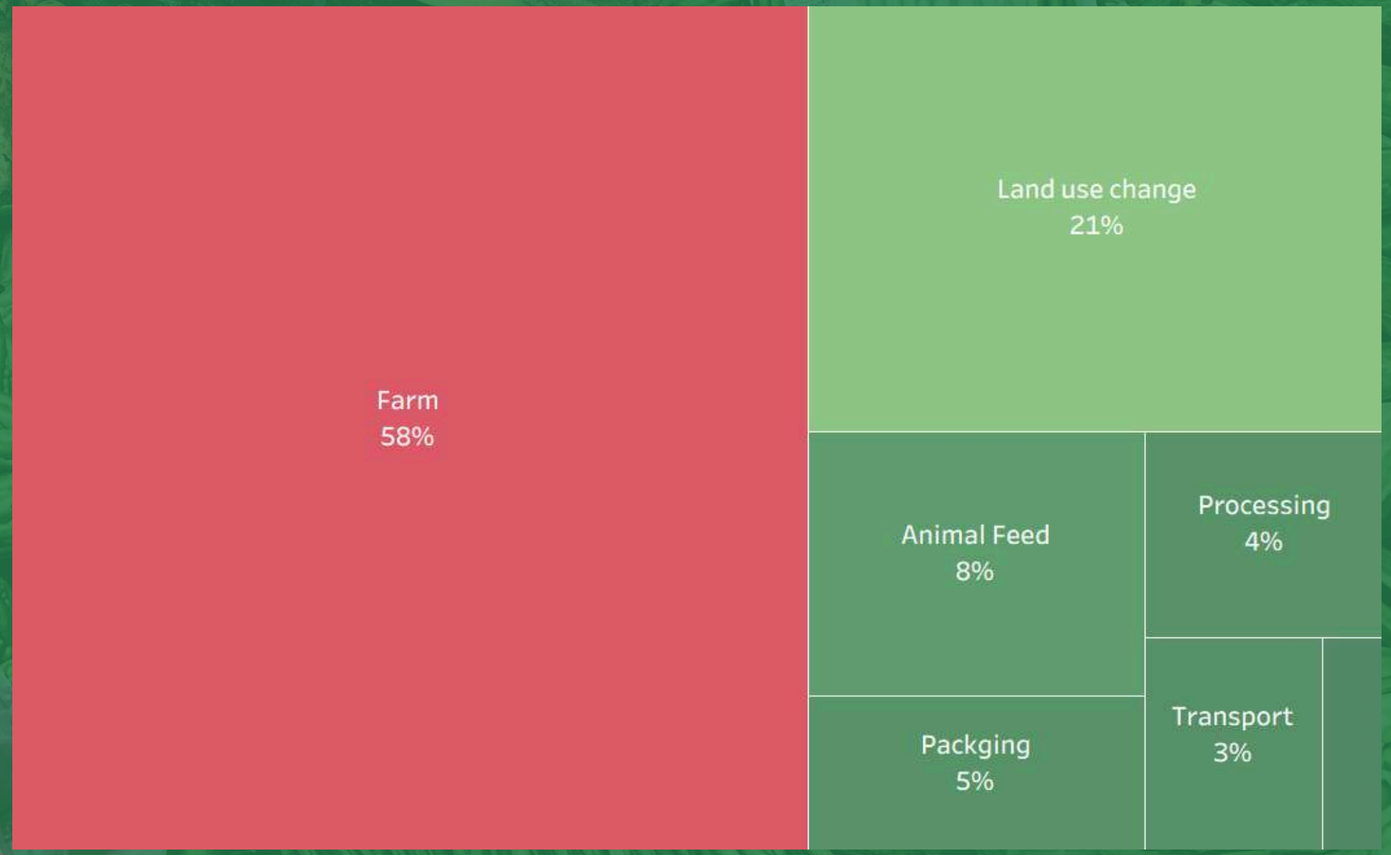
For most foods, what you eat is far more important than where it comes from.

Choosing low-impact foods is much more effective than choosing local high-impact foods.

For most foods, over 80% of emissions come from land use change and farm-stage processes.

Transport typically represents less than 10% of total food emissions,

and for beef it is often below 1%.

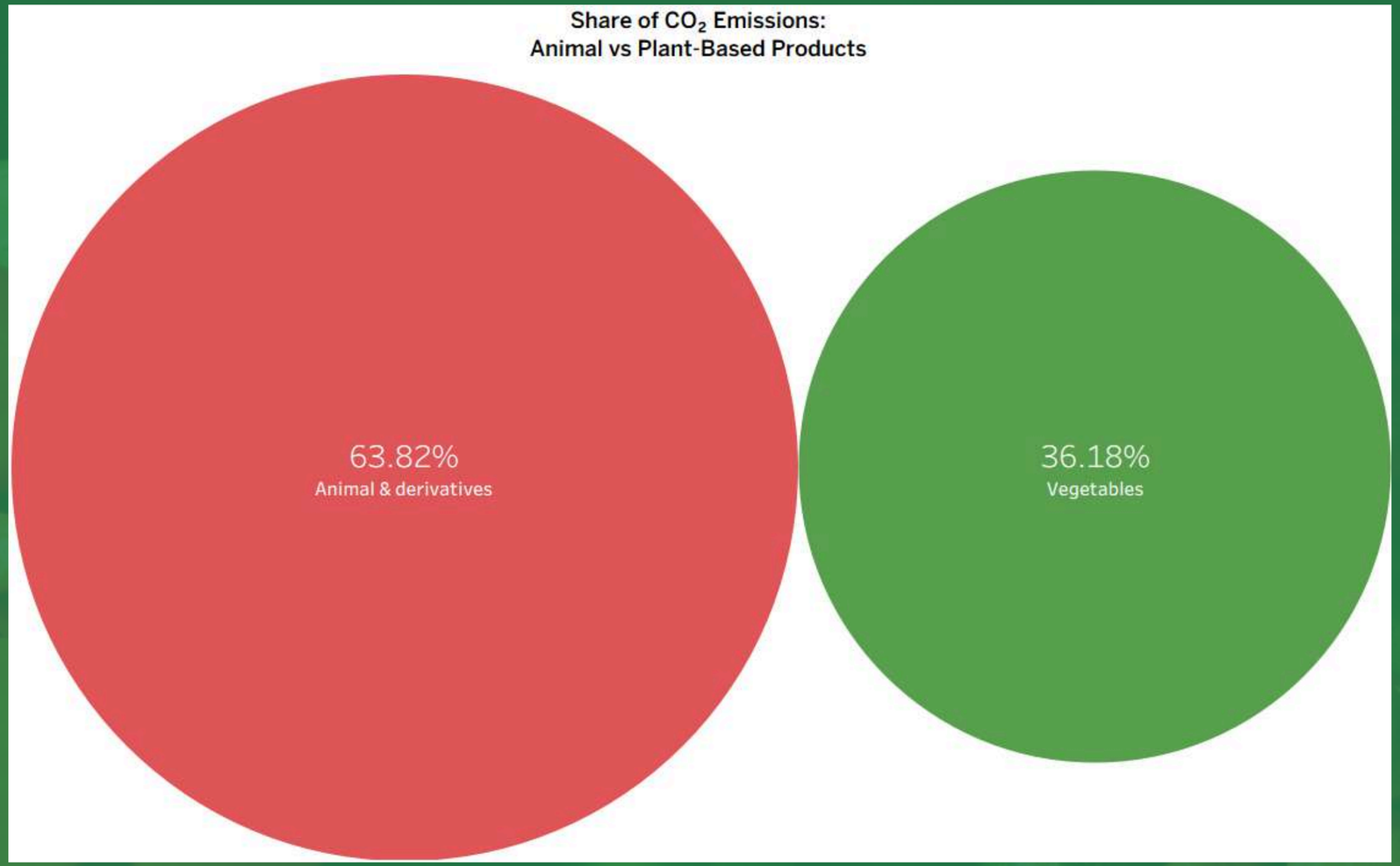


ANIMAL-BASED FOODS DRIVE MOST EMISSIONS

1 Animal-based foods account for the majority of food-related greenhouse gas emissions.

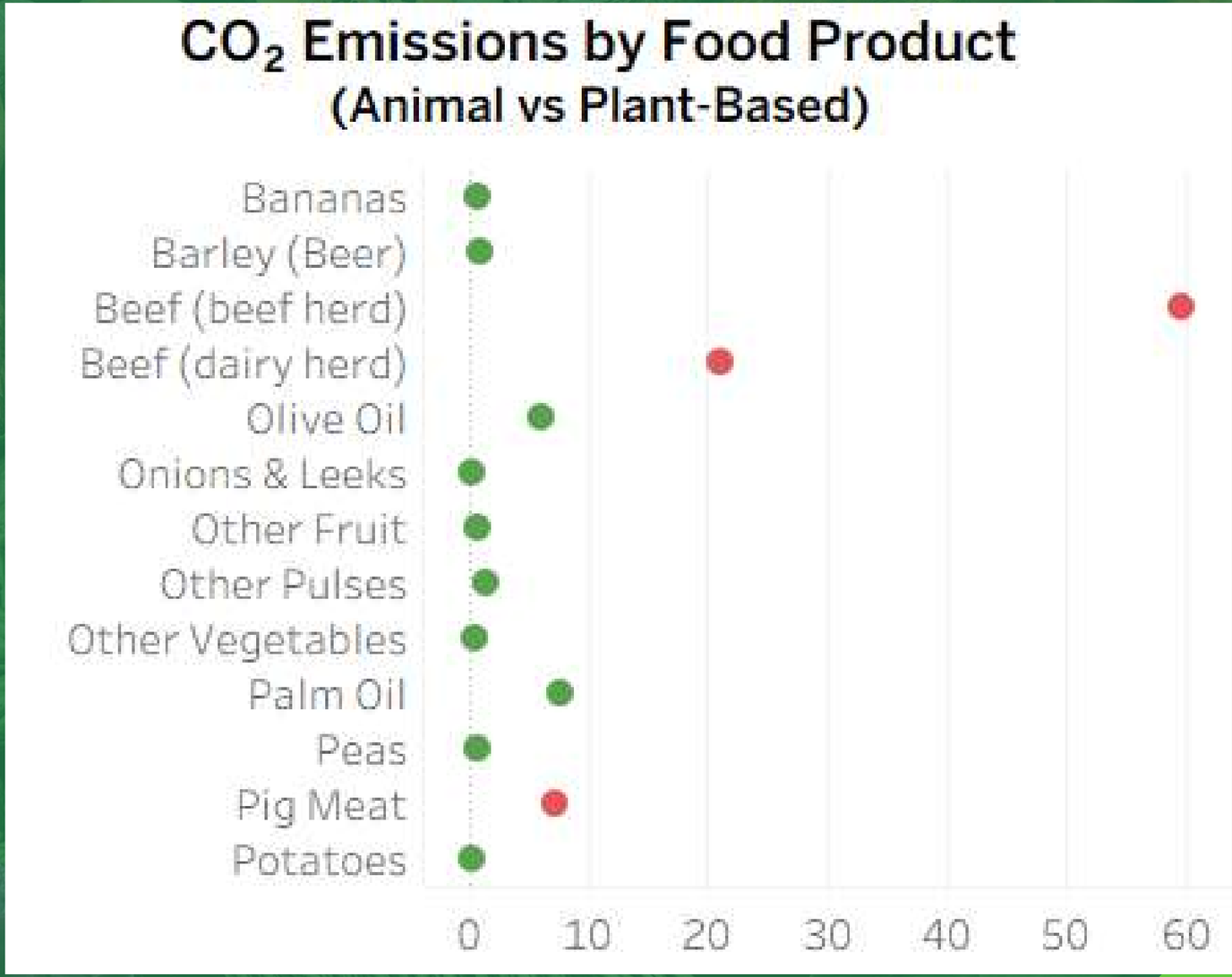
2 Plant-based foods consistently show much lower carbon footprints.

3 This makes menu composition a critical lever for sustainability.



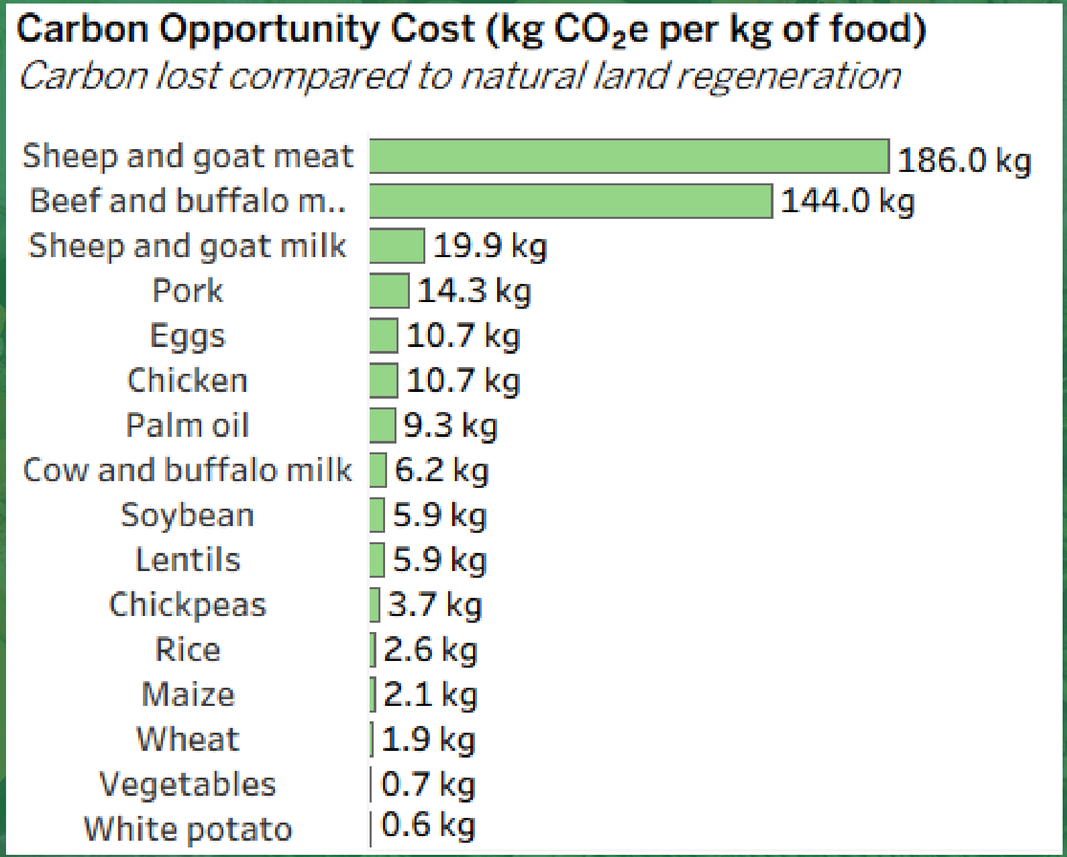
MASSIVE DIFFERENCES BETWEEN FOODS

The carbon footprint of food products varies enormously. Producing **1 kg** of **beef** emits around **60 kg** of **CO_2e**, while **peas** emit about **1 kg** of **CO_2e**. The difference between food types is far greater than the difference between local and imported food.



THE HIDDEN COST OF LAND USE

Animal-based foods, especially beef and lamb, prevent large amounts of carbon from being stored in natural ecosystems.



This "opportunity cost" means their real climate impact is even higher than direct emissions alone.

WHAT THIS MEANS FOR MENU DESIGN

Reducing red meat offers the largest carbon reduction

Increasing plant-based dishes delivers strong climate benefits

Sustainability gains come mainly from what is served, not from where it is sourced

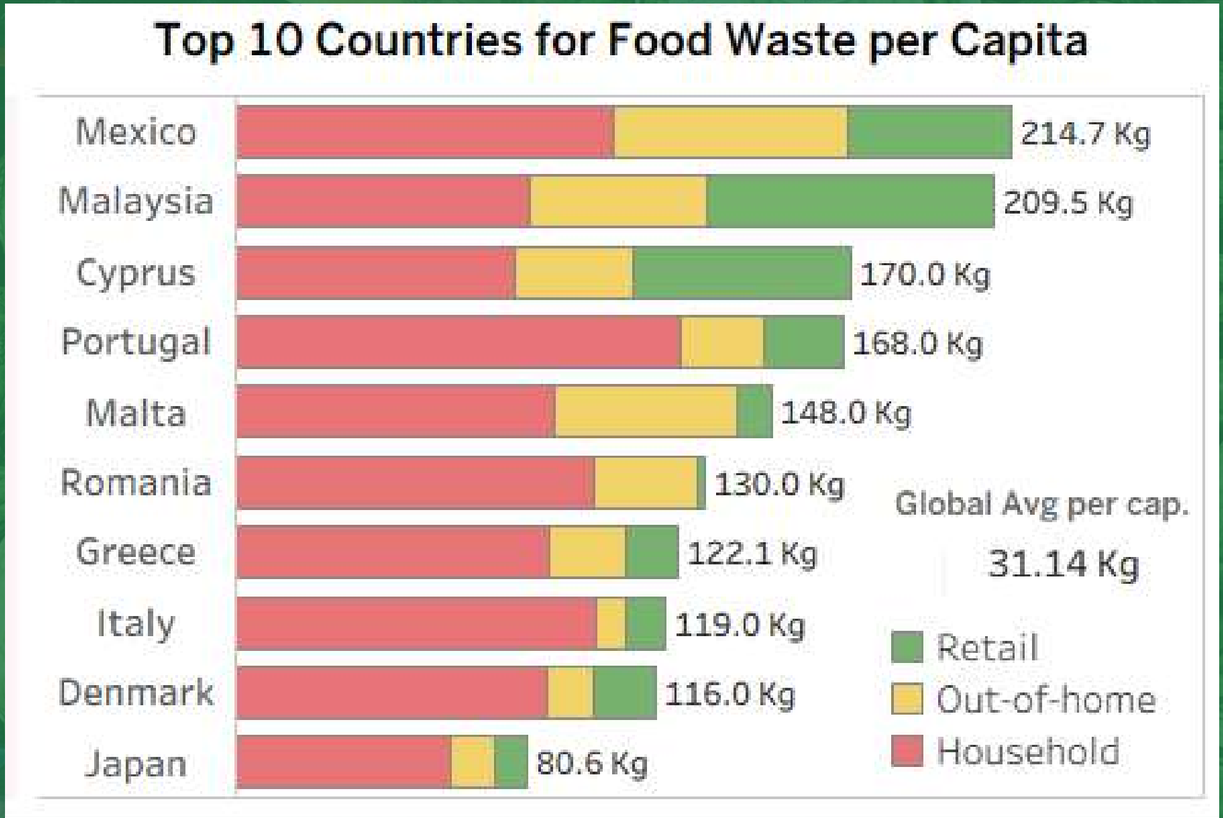
Replacing beef and lamb with poultry or plant-based proteins drastically lowers emissions

REDUCE FOOD WASTE, REDUCE CO_2

Food waste is not just a social issue – it is also a climate issue.
When food is wasted, all the emissions generated to produce, process, and transport it are wasted too.

Call to Action

- Plan meals and portions carefully
- Prioritize plant-based options
- Store food correctly to avoid spoilage
- Use leftovers creatively
- Make low-carbon choices visible on menus



LOW-CARBON MENU STRATEGY

- Make plant-based dishes the default option
- Limit high-impact items (beef, lamb, high-emission dairy)
- Promote low-carbon alternatives on the menu
- Educate customers with simple labels (“Low-Carbon Choice”)
- Maintain taste and attractiveness to ensure adoption

CONCLUSION

Food choices are one of the most effective ways to reduce climate impact.
Data shows that changing what we eat matters far more than eating local.

A restaurant can significantly reduce its carbon footprint by shifting menu composition toward plant-based and low-impact foods – without sacrificing quality or customer experience.

INTERACTIVE DASHBOARDS AVAILABLE ON TABLEAU PUBLIC (LINK IN PRESENTATION)